5
Engineering Scale-up for Hydrogen Transport Membranes

David J. Edlund

There has always been a special challenge to engineering practical membrane modules for commercial and industrial applications. Without successful incorporation into a membrane module, membranes are virtually useless – relegated to laboratory experimentation but not of practical utility. For clarity, the term membrane refers to the permselective layer: a dense, thin metal foil or thin metal layer that is deposited onto a porous support. The term membrane module, or simply module, refers to the housing for the membrane, inclusive of one or multiple membranes. Significant physical features of the membrane module housing include the feed and permeate flow channels, seals that isolate the feed and permeate flow channels, and the external shell, which may in fact be a high-temperature pressure vessel, depending on the design operating temperature and pressure for the module. It is very typical for dense, hydrogen-permeable metal membranes to be used at temperatures >300 °C (and frequently as high as 500 °C to 600 °C) and feed pressures >7.93 bara (115 psia or 100 psig).

An example of a commercial membrane module is shown in Fig. 5.1. This membrane module utilizes multiple foil membranes of Pd–40Cu (40 wt% Cu) to



Figure 5.1 Photograph of a commercial membrane module containing a total membrane area of approximately 3700 cm^2.

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separate high-purity hydrogen from reformate in a single step. (Reformate is the term given to the product stream from a steam reformer, partial-oxidation reformer, or autothermal reformer. The composition of reformate is predominantly CO_2, CO, H_2, CH_4, H_2O, and various other components at low concentrations.) The external shell of the membrane module is designed to be operated at pressures up to 200 psig (14.8 bara) and temperatures up to 450 °C.

This chapter will address the special considerations that apply to incorporating dense, hydrogen-permeable metal membranes into practical membrane modules for commercial and industrial use. It is organized to present a brief historical overview, a general review of hydrogen-permeable metal membranes, scale-up from laboratory test-and-evaluation membrane modules to commercial membrane modules, membrane module design and construction, and commercial applicability.

5.1 Historical Review

In the mid-1950s, Union Carbide conducted several pilot-scale tests of palladium alloy membranes for separating and purifying hydrogen from various process streams. Tubular membranes were used in cylindrical modules for these tests. Unfortunately, the poor durability of the brazed membrane seals eventually led to termination of the work. The first successful commercial hydrogen purifier is often credited to J. Bishop & Company Platinum Works (Malvern, Pennsylvania). A 1960 report by J.B. Hunter [1] describes the successful implementation of a palladium–silver alloy fabricated as thin tubes. Although the exact alloy composition is not disclosed, it is very likely that the silver content is in the range of 23 to 25 wt%. Hunter does not provide details about the module design, but he does provide a photograph of a membrane tube bundle. This membrane module design was used in some industrial applications including purifying 3260 $Nm^3\ d^{-1}$ of hydrogen at the National Cylinder Gas Division of the Chemetron Corporation (Los Angeles, California, 1962) [2] and purifying 42 $Nm^3\ h^{-1}$ of hydrogen from dissociated ammonia at Kreisler Industries Corporation (East Paterson, New Jersey, 1969) [3].

By 1969, Johnson Matthey had acquired the palladium membrane technology from J. Bishop & Company Platinum Works and was working diligently on commercial applications of the technology. One of the earliest reports [4] on membrane module design, and the importance of module design for the performance of the hydrogen purifier, addresses issues of mass transfer at the feed side of the membrane as well as adequacy of heating the incoming feed stream. A crude form of a planar membrane module is discussed, but for the most part the report focuses on tubular membranes in a cylindrical shell. The membranes are brazed to a header at one end only to allow for expansion along the length of the tubular membrane. A photograph of these cylindrical membrane modules indicates that they were generally long relative to the diameter, the length being about eight to twenty times the diameter [5].

Today, interest in dense-metal, hydrogen-permeable membranes remains strong for a variety of commercial applications ranging from fuel cells to coal gasification. This interest is divided between tubular membranes and planar membranes. Module designs are similar to those developed for polymeric membranes with the exception of the spiral-wound polymeric-membrane module for which there is no analogous design under development using dense-metal membranes. Although many groups worldwide are developing new membrane compositions, fewer are working to develop membrane modules for small-scale to large-scale commercial applications (e.g., Power + Energy, tube-and-shell membrane module; IdaTech, planar membrane module; and Hy9, planar membrane module).

5.2
General Review of Hydrogen-permeable Metal Membranes and Module Design

For reasons of convenience and economy, laboratory-scale test-and-evaluation membranes are often relatively small in size, less than 5 cm^2 (1 inch diameter) being quite common and in some cases as small as 0.3 cm^2 to 0.4 cm^2. These test-and-evaluation membranes are almost always circular in shape and sized to fit into a permeation test cell. Membranes that are only 0.3 cm^2 to 0.4 cm^2 in area are sized to fit into a standard $^1/_4$ inch (6.35 mm) commercial face-seal compression fitting. Although convenience figures prominently in the design of laboratory test apparatus, it is not a dominant design criterion for commercial and industrial applications where economics and durability are of primary importance.

Commercial and industrial applications are discussed in greater detail below, but include on-site hydrogen production and purification (electrolysis and reforming) for laboratory instruments, metals production, generator cooling, and alternative energy technology, principally fuel cells. It is important to understand that practical applications will require from about 100 cm^2 to thousands of square meters of membrane area. This is an exceedingly large range, and it stands to reason that a single membrane module design may not serve well over this entire range of membrane area. However, the underlying principles that guide the design of the membrane module are equally applicable at both ends of the range.

As discussed elsewhere in this text, there are two types of dense, hydrogen-permeable metal membranes to consider from the perspective of module scale-up and design: thin metal foils and permselective metal layers formed on a porous support. Another class of hydrogen-permeable inorganic membranes – dense proton-conducting ceramic membranes – are still under development and are addressed in Chapter 2.

As with any engineering exercise, when the objective is to successfully scale-up and develop a product for commercial and industrial applications, the focus must be on achieving acceptable economics. Of course, economics involves the upfront capital expenditure, as well as any ongoing maintenance costs and fuel or other utility costs. One of the potential advantages of dense metal membranes for hydrogen purification is that the membranes are preferably operated at signifi-

cantly elevated temperatures – often 300 °C to 600 °C. Assuming that the feed stream is generated at high temperature (as is the case for reforming and partial oxidation chemical processes) then there should be no cost associated with heating the membrane module because the hot feed stream will suffice for this purpose. However, if the hydrogen-rich feed stream is a low-temperature by-product or waste stream, or the output from an electrolyzer, then there may be some fuel cost associated with heating the membrane module and feed stream.

5.2.1 Scale-up and Differential Expansion

So, what are the challenges facing engineers in the transition from laboratory-scale membrane permeation cells to commercial, large-area membrane modules? Quite simply the answer is scale. As the membrane area is increased, differential expansion between the membrane and the membrane module becomes a very significant design factor. Differential expansion between the membrane and the membrane module may be due to differing coefficients of thermal expansion and/or hydrogen-induced expansion of the metal membrane (i.e., the permselective layer). Table 5.1 compares the coefficients of thermal expansion (CTE) at 298 K for Pd–40Cu, Pd–23Ag (two common hydrogen permeable membrane alloys) and many likely structural metals for use in fabricating membrane modules.

To illustrate the importance of scale and the coefficient of thermal expansion, let us analyze two membrane modules that are assembled at room temperature (nominally 25 °C) and then heated to 400 °C for operation. One membrane mod-

Tab. 5.1 Coefficients of thermal expansion at 298 K for membrane and membrane module structural materials [6].

Alloy type/Grade	CTE ($\mu m\ m^{-1}\ °C^{-1}$)
Pd–40Cu	14.9
Pd-23Ag	12.6[a]
Monel 400 (UNS N04401)	13.9
Type 304 Stainless Steel (UNS S30400)	17.3
Type 316 Stainless Steel (UNS S31600)	16.0
Type 661 Stainless Steel (UNS R30155)	14.0
Inconel 625 (UNS N06625)	12.8
Haynes 230 (UNS N06002)	12.7

a A.C. Bailey, N. Waterhouse, B. Yates, The thermal expansion of palladium–silver alloys at low temperatures, *J. Phys. C: Solid State Phys.* **1969**, *2*(2), 769–776.

ule is a common single-membrane permeation cell used for laboratory test-and-evaluation experiments; the other membrane module is a commercial design (modified for this example) consisting of 64 planar membranes. These two membrane modules are shown in Fig. 5.2. Both membrane modules use flexible graphite gaskets to seal the membranes (thereby isolating the feed and permeate channels). Both membrane modules use 304 stainless steel, and both modules use 304 stainless steel bolts with spring washers to compress the gaskets and to secure the end plates (see also Fig. 5.9). For each membrane, there are two gaskets: one associated with the feed side of the membrane and the other associated with the permeate side of the membrane. A typical gasket thickness is 0.01 in (0.254 mm) prior to compression. The gaskets will normally be compressed by 30% such that, after assembly of the membrane module and tightening the bolts, the gaskets are 0.007 in thick (0.178 mm).

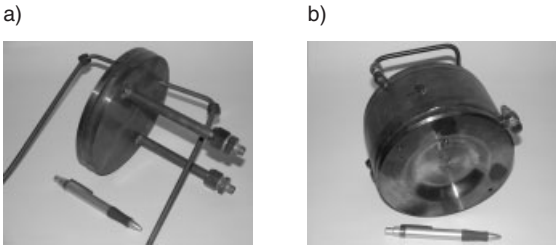


Figure 5.2 A single membrane module for laboratory evaluations (A) and a commercial membrane module containing approximately 3700 cm^2 total membrane area (B).

The coefficient of thermal expansion of 304 stainless steel is 17.3 μm m^{-1} $°C^{-1}$, so, as the membrane modules are heated from 25 to 400 °C, the laboratory-scale membrane module will expand about 13 μm in length while the commercial-scale membrane module will expand about 460 μm. Although the ratio of the change in length $\Delta l/l$ remains the same for both membrane modules, and is defined as the coefficient of thermal expansion of 304 stainless steel, the actual magnitude of Δl is quite different for the two modules. Since this expansion in the overall length of the membrane module will result in a relaxation of the compressive force on the gaskets, there is the potential for leaks to develop – leaks that were not present at room temperature when the module was assembled, but appear only when at or near to the design operating temperature. It is important to realize that a leak may not form initially, but as the membrane module is cycled through repeated temperature swings from 400 to 25 °C and back to 400 °C, the gaskets may fatigue – that is, lose resiliency – resulting in the appearance of leaks at a later time as the module is aged.

However, relaxation of gasket seals is not the only way that differential expansion can adversely affect the performance of membrane modules. It is important to remember that each membrane in the module is constrained by the module construction. If the membrane is in the form of a planar, flat sheet, then it is con-

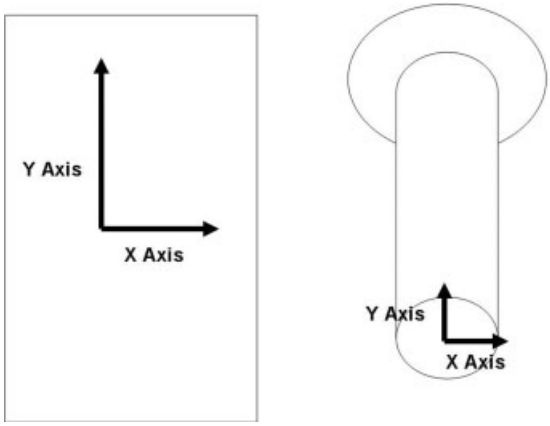


Figure 5.3 Schematic representations of expansion axes for planar and tubular membranes.

strained in two dimensions (the x and y axes). If the membrane is a tube, fixed to a header at one end and free floating at the other, then it is also effectively constrained in two dimensions (the x and y axes). This is shown schematically in Fig. 5.3. Even worse would be a tubular membrane that is fixed at both ends to headers, resulting in a membrane that is constrained in three dimensions. Surprisingly, the latter is a design that is still often used in laboratory test-and-evaluation cells using tubular metal membranes (usually a thin permselective layer deposited onto a porous tube).

Now, we will consider two different cases of mismatch in thermal expansion of the membrane (the permselective layer) and the membrane module. In the first case, it is assumed that the membrane module expands to a greater degree with increasing temperature than does the membrane, while in the second case we will assume that the membrane module expands to a lesser degree with increasing temperature than does the membrane. The first case will result in a tensile force being applied to the membrane during heating cycles. This tensile force will either tear the membrane, resulting in immediate failure, or will stretch it. If the membrane is only stretched, then as the membrane module is cooled during thermal cycling, the membrane will be subjected to a compressive force, causing the membrane to wrinkle. Repeated thermal cycling will result in the membrane being pulled to its maximum dimensions (at elevated temperature) and then pushed to its minimum dimensions (at reduced temperature). This repeated physical distortion of the membrane will lead to fatigue failure.

Analysis of the second case, in which the membrane module expands to a lesser degree with increasing temperature than does the membrane, yields the same result, although the membrane is not subjected initially to a tensile force. The membrane will still be subjected to repeated physical distortions ultimately leading to fatigue failure. Soft seals, such as flexible graphite gaskets, are somewhat accommodating of differential expansion, whereas hard seals (brazed or welded) are not.

Although mismatch of thermal expansion is a well known issue in engineering, the expansion of the metal membrane due to hydrogen dissolution may be far greater than any thermal expansion. This is the case for Pd–23Ag, which has been measured to swell as much as 3% when equilibrated under 7 bara hydrogen at 400 °C. Other palladium alloys, such as Pd–40Cu, do not swell to such a large degree in the presence of hydrogen. These results are consistent with the solubility of hydrogen in each alloy; hydrogen dissolves to a greater degree in Pd–23Ag than it does in Pd–40Cu. A straightforward experiment to test the cyclic durability of a candidate membrane under hydrogen swelling conditions only (not thermal) is the following. Place a sample membrane, at least 5 cm^2 in area, in a suitable permeation test cell and heat the membrane in the range 300 to 400 °C. Now, equilibrate the membrane under 7 bara of pure hydrogen on the feed side and measure the hydrogen flux. There should be a slow flow of hydrogen through the membrane module, and the flux should be constant. Without changing the feed pressure, switch the feed gas from 7 bara hydrogen to 7 bara nitrogen. (This is easily done by connecting the regulated output from a cylinder of compressed hydrogen to a ball valve, and then connecting the output of the ball valve to one branch of a tee. The regulated output from a cylinder of compressed nitrogen is also connected to a second ball valve, and the output to the second branch of the tee. Finally, the third branch of the tee is the feed line to the membrane permeation cell. By alternately opening and closing each ball valve, the feed is abruptly changed from hydrogen to nitrogen and back again.) A plug of nitrogen will pass into the membrane module, and the flux of hydrogen will cease. Because the temperature remains unchanged, this abrupt change in hydrogen partial pressure stresses the membrane, but only due to hydrogen-induced lattice expansion. After many cycles, perhaps 100, the membrane will fail if the degree of hydrogen-induced expansion is significant. Pd–23Ag membranes fail this test; Pd–40Cu membranes have passed 1000 cycles without failure.

When there is a significant mismatch in expansion of the membrane relative to the module materials, wrinkles will appear in the membrane. Figure 5.4 shows two planar membranes of Pd–40Cu that have been repeatedly cycled from 20 to 400 °C and back to 20 °C. In one case the coefficient of thermal expansion of the

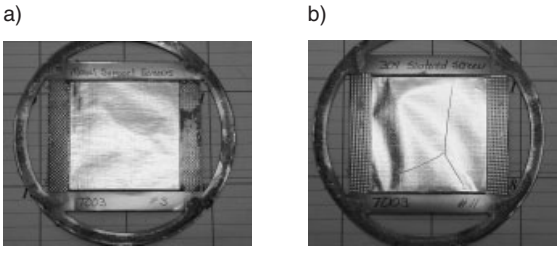


Figure 5.4 Comparison of effects on a palladium-copper alloy membrane by thermal cycling: (A) a Monel membrane support, (B) a 304 stainless steel membrane support.

Pd–40Cu was approximately matched by using Monel in the module components in direct contact with the membrane (the membrane support), whereas in the other case 304 stainless steel was used for the membrane support. As shown in Table 5.1, the coefficient of thermal expansion of 304 stainless steel is substantially greater than the coefficient of thermal expansion of Pd–40Cu. Failure occurred after only a few cycles in the case of the 304 stainless steel components, whereas the Monel components demonstrated much improved durability.

To demonstrate that flexible graphite gaskets can accommodate some difference in the degree of expansion between the membrane and module components, the experiment was repeated using feed-channel and permeate-channel frames made of 304 stainless steel, while the membrane support remained Monel. Cyclic durability was very good, and the membranes remained devoid of visible wrinkles.

Given that the membranes and modules are often complex engineered structures, it is not immediately obvious if design problems exist, and, therefore, careful consideration is necessary. For instance, a very thin dense metal membrane must be supported by an underlying porous (or other) structure if the membrane is to be used under significant pressure differentials. This is true for thin metal foils as well as for thin permselective metal layers that are deposited onto a porous support (such as by vapor deposition or plating processes). In the latter case, it is important that the coefficient of thermal expansion of the membrane and of the porous support be similar – a large mismatch will result in membrane failure after a relatively few start–stop cycles.

In all of the aforementioned examples, it is cycling of the membrane module that places the maximum degree of stress on the design. It is possible, if not likely, that membrane modules which are generally of poor design will survive long periods of operation if the module is not thermally cycled. However, such is not often the case, and practical consideration must be given to the need to thermally cycle membrane modules for most, if not all, applications of commercial interest.

5.2.2
Overview of Sealing Methods

Matching components to minimize differential expansion is not the only design consideration of importance when scaling-up membrane modules. Fluid flow dynamics through the membrane module must also be considered. Fluid flow is dominated by two design criteria: pressure drop and mass transfer, both of which are usually improved by increasing the feed-side pressure. However, this must be balanced with the need to support very thin membranes against large pressure differentials and the need to engineer the membrane module shell (or housing) to contain the pressurized feed stream at the design operating temperature for the module. The module shell rapidly becomes very heavy and expensive when feed pressures become several hundred psi (14.5 psi = 1 bar) and operating temperatures are 400 to 600 °C. A subtle, but significant, drawback of an overly heavy membrane module is that there is significant thermal time lag on heating the membrane module to the operating temperature.

Membrane-to-module sealing methods are very important and have been addressed briefly above. Indeed, the functionality of the membrane module is predicated on the assumption that the feed stream is isolated from the permeate stream and the external atmosphere, and that the permeate stream is also isolated from the external atmosphere. This is especially true for a membrane module that will be used to purify hydrogen, since leaks could result in fire or explosion hazards.

There are three different sealing methods that have been used to varying degrees of commercial success. They are:

- Gaskets, flexible graphite being preferred, but also including soft metal seals such as copper. Volatile metals and metals which have a low melting point must be avoided (e.g., tin, lead, zinc, cadmium, indium).
- Brazing, usually involving a conventional braze alloy in which application involves forming the liquid phase of the braze alloy. As with soft metal gaskets, volatile components in the braze alloy (including phosphorus) should be avoided. A special subset of brazing is solid-state brazing, or intermetallic diffusion. This occurs spontaneously, albeit slowly, when miscible metals are placed in direct contact. Examples of suitable metals for solid-state brazing to the membrane and module include copper, iron, silver, and nickel [7].
- Welding, difficult to achieve on a large scale and may require specialized micro-welding techniques that minimize the heat flux into the weld zone.

Laboratory membrane modules for test-and-evaluation experimentation almost always utilize gasket seals to facilitate removal of the membrane after testing and reuse of the permeation test cell. Gaskets have also been extensively used in assembling larger commercial membrane modules using planar membranes, but brazing has historically been the preferred sealing method for small diameter tubular membranes used in commercial applications.

5.3 Scale-up from Laboratory Test-and-Evaluation Module to Commercial Membrane Module

Laboratory permeation test cells must be reusable. A commonly used design for testing thin metal foil membranes is shown in Fig. 5.5. It makes use of simple and inexpensive gaskets to seal the membrane, and the membrane is supported on a sintered metal disc, a porous ceramic disc, or a metal screen disc. Other designs of laboratory permeation cells have been reported [8, 9]. Although these types of permeation test cell are very useful for pure gas testing, they are not so useful for conducting experiments in which high recovery of hydrogen is desired

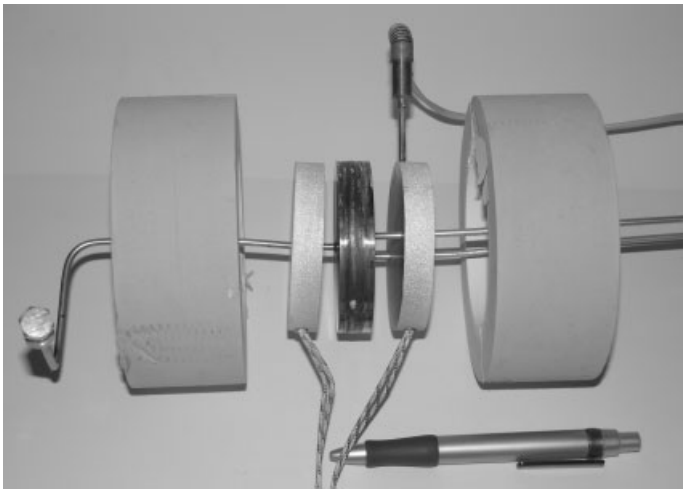


Figure 5.5 A laboratory module, with insulation and electric heating elements, for evaluating single membranes.

from a mixed gas feed stream for reasons of nonuniform and generally poor flow distribution.

The surface condition, porosity, tortuosity, and thickness of the support structure are all important features. Clearly, the surface condition of the support must be such that the membrane that is resting on the support structure is not physically damaged. Generally, this implies that the surface of the support must be smooth. Also, there should be very little resistance to flow of the permeate hydrogen through the support structure. This requirement favors highly porous, straight-through channels, and a relatively thin structure. Wire mesh screens are very good choices for meeting all of these requirements, whereas sintered metal often cannot meet the requirement for a smooth surface, and ceramic supports are generally relatively thick with highly tortuous channels. The laboratory permeation test cell shown in Fig. 5.5 will frequently use flexible graphite gaskets. While these are convenient, readily available, and can be easily cut from sheet stock, graphite does slowly oxidize in air at temperatures ≥400 °C, limiting the length of time that the permeation cell can be tested. This drawback limits the duration of lifetime testing. At very high temperatures (600 to 700 °C), depending on the dimensions of the gasket and the exposed surface area of the gasket edge, it may be possible to operate only for days to a few weeks before the gasket is oxidized and the seal compromised.

As we shift our interest from laboratory permeation cells to commercial membrane modules, our focus shifts from one of convenience to one of economics. The initial cost, or capital cost, of the membrane module is very important, but we cannot ignore the ongoing operating costs which include any and all maintenance, as well as dedicated energy and utility costs. As mentioned previously, if the feed stream to the membrane module is sufficiently hot and supplied at the

operating pressure of the module, the dedicated energy and utility costs will be zero. Because the permeation of hydrogen through the membrane is a thermally neutral process, it is not necessary to either add heat or remove heat from the membrane module during operation. However, if the feed stream needs to be either compressed or heated, then the energy costs may be substantial.

For a well designed membrane module of reasonably large scale, the initial cost should be dominated by the cost of the palladium content of the thin metal membrane (assuming a palladium alloy comprises the permselective layer). The module itself will be made of steel, most likely a nickel alloy, with or without significant chromium addition. The cost of the steel and the assembly labor should not exceed the cost of the palladium alloy membrane.

It is often found that the sintered metal and porous ceramic supports that have been used for many academic membrane studies are very expensive, sometimes even exceeding the cost of the palladium alloy membrane by several fold. This situation cannot be accepted for commercial membrane modules, especially when a large membrane area of up to several hundred to several thousand square meters is required. Some of the least expensive membrane supports include tension springs for tubular membranes and woven wire mesh for planar membranes, but even these supports can be costly, and further development of exceptionally low cost membrane supports is needed.

5.3.1 Cost and Membrane Thickness

A very powerful strategy for reducing the cost of membrane modules, assuming that the cost is dominated by the cost of the palladium (or other metals) in the permselective membrane, is to reduce the thickness of the membrane. To illustrate this point, consider a membrane of sufficient area, A in m^2, to purify 1 $Nm^3\ h^{-1}$ of hydrogen from the feed stream. It is assumed that the membrane module is performing ideally, meaning that there are no significant mass transfer resistances at the feed side of the membrane and no significant pressure drop through the support at the permeate side of the membrane. For the purposes of this example it does not matter under what conditions the membrane is operated, nor what the feed composition is – only that both are held constant for the analysis. It is further assumed that the membrane is an alloy of palladium (e.g., Pd–23Ag or Pd–40Cu), that the permselective layer is 25 μm thick, and that the required cost of the palladium content for a unit area of the membrane is valued at U.S.$2000 based on 2005 market prices.

Now, consider that the membrane thickness is reduced to 10 μm, thereby reducing the mass of palladium per unit area of membrane. Assuming that the initial 25 μm thick membrane of unit area contained US$2000 worth of palladium, the new 10 μm thick membrane will contain (US$2000)(10 μm/25 μm) or US$800 worth of palladium. The hydrogen flux will increase by the ratio 25 μm/10 μm or 2.5. Because the flux increases by a factor of 2.5, the total membrane area will be reduced by the same factor in order to keep the permeate hydrogen flow rate con-

stant at 1 $Nm^3\ h^{-1}$. The net result is that the total value of palladium represented by the new 10 μm thick membrane, sized to deliver 1 $Nm^3\ h^{-1}$ of permeate hydrogen, is US$800/2.5 = US$320. Thus, if it is assumed that the membrane material costs dominate the module cost, and if it is further assumed that the flux of hydrogen through the membrane is dominated by the membrane thickness, then for a fixed delivery rate of product hydrogen, the cost of the membrane (and membrane module) will be reduced according to the following expression:

$$\text{cost reduction factor} = (\text{initial membrane thickness}/\text{reduced membrane thickness})^2 \quad (5.1)$$

where the term "cost reduction factor" is the factor by which the reduced thickness membrane is less expensive than the initial, thicker membrane. In the above example, the cost reduction factor is $(25/10)^2$ or 6.25; and dividing this factor into the initial cost of US$2000 gives us the final reduced cost of US$320.

The data presented in Fig. 5.6 were derived from membrane modules using Pd–40Cu foil (planar) membranes with the membrane area sized to deliver 0.78 $Nm^3\ h^{-1}$ of product hydrogen. Even though palladium alloy membranes are often criticized as being too expensive for commercial applications due to the cost of palladium, it is clear that as membrane thickness is reduced to 10 μm and less, the cost associated with the value of palladium in the membrane can be very reasonable.

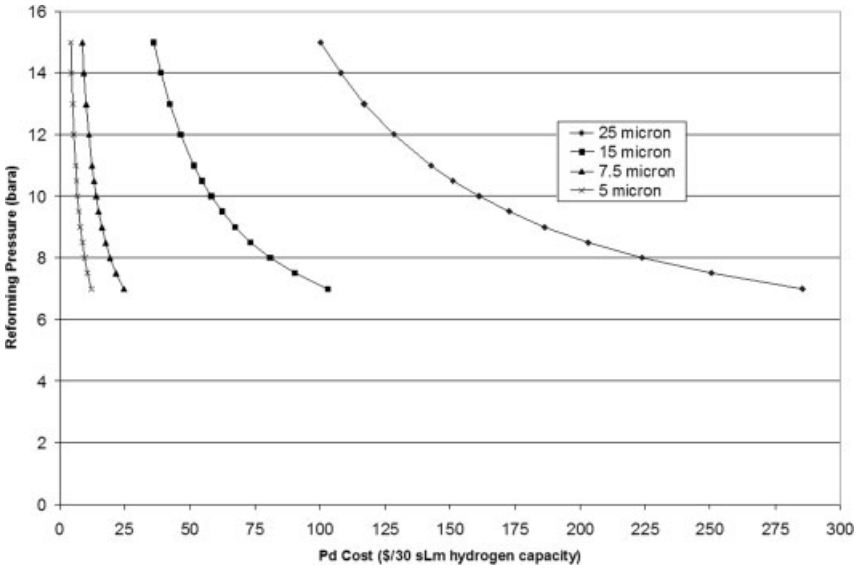


Figure 5.6 The dependence of hydrogen flux and membrane cost on membrane thickness. The feed contains 53% hydrogen, and hydrogen recovery is 70%.

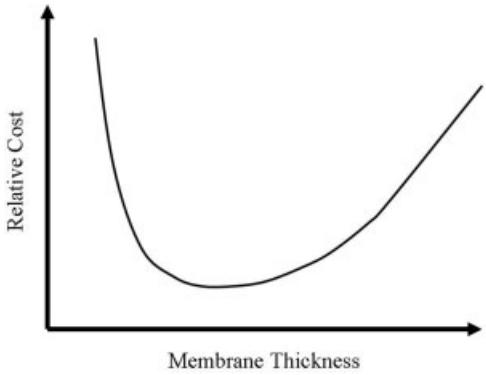


Figure 5.7 Qualitative graph of cost versus membrane thickness showing a cost minimum with membranes of intermediate thickness.

So, if thin membranes are a good idea, what about ultra-thin membranes that are only 1 or 2 μm thick? Unfortunately, fabrication and handling costs do not decrease linearly with reduction in thickness of the permselective membrane. Experience has shown that qualitatively the relationship between fabricated cost and membrane thickness is approximately as shown in Fig. 5.7. Initially, as the thickness of the membrane is reduced, there is a corresponding reduction in the cost, largely driven by a reduction in the mass of expensive palladium used in the membrane. However, as the membrane thickness is reduced to approximately <5 μm, the fabrication cost increases faster than can be offset by the reduction in the amount of palladium. A major reason that fabrication costs increase so dramatically is the reduction in the yield of satisfactory membrane product (i.e., membrane product without pinholes, tears, and similar defects). The membrane must be free of defects which would compromise the permselectivity of the membrane. This requirement becomes increasingly difficult to satisfy as membranes become exceptionally thin.

Also, as membranes, whether tubular or in the form of a flat membrane foil, become exceptionally thin (<5 μm), the handling of the membranes becomes increasingly difficult, leading to additional expense in the fabrication of the membrane modules. Also, the probability of damaging membranes during assembly of the modules increases, resulting in yield declines and corresponding cost increases.

If the permselective membrane is a thin metal layer deposited onto a porous support, the challenge of making defect-free metal coating increases as the thickness of the layer is reduced. This challenge is compounded by the fact that the underlying support is porous and rough. The net effect is reduced yield of acceptable membranes and increased cost of fabricating membrane modules.

5.3.2
Module Maintenance and Operating Costs

A hydrogen-separation membrane module will not require any maintenance provided that:

- The feed stream entering the membrane module is chemically compatible with the materials used to fabricate the membrane and module.
- The feed stream is filtered to prevent accumulation of particulates within the membrane module.
- The membrane does not fail during the design life of the module.

The need for chemical compatibility and removal of particulates by filtering are fundamental requirements that should be anticipated during the module design process. A large body of information is available to guide proper materials selection, and there are many adequate filter means commercially available.

Membrane failure modes have been discussed above, and the connection to module design has also been discussed. Poor design for cyclic durability will be measured by the customer – the operating costs will be adversely affected by the requirement to replace membranes prematurely. Although the cost of membrane replacement should be offset by a recycle credit, the cost of materials and labor, and potential lost productivity, is still likely to be significant. The credit for recycling palladium alloy membranes may be as great as 95% of the market value of the palladium (for foil membranes, perhaps only 85% of the market value if the palladium alloy is deposited onto a porous substrate). Environmentally, recycle also offers benefits versus recovery and purification of palladium from ore.

5.3.3
Overview of Membrane Fabrication Methods

The most common methods for manufacturing thin metal membranes include rolled foil, drawn tubes, and films deposited onto porous substrates (ceramic or sintered metal). Usually, electroless plating or electrolytic plating are the methods used to deposit the permselective metal onto the porous substrates although vapor deposition methods have been the subject of much research effort. However, to date, vapor deposition methods have not proven to be a superior membrane fabrication method. There are pros and cons to each of these methods, but commercial membrane modules have only succeeded using rolled foil and drawn tubular membranes.

The physical properties of the specific metal alloy used for the hydrogen permeable membrane strongly influence the sequence of steps required to roll foil and draw thin-walled tube. If the alloy is either too hard or too soft, it will be difficult to work; if the alloy work-hardens quickly, frequent annealing steps will be necessary. Pure palladium, Pd–23Ag, and Pd–40Cu have all been fabricated into thin foil and drawn tubes. The palladium alloys are somewhat easier to fabricate into

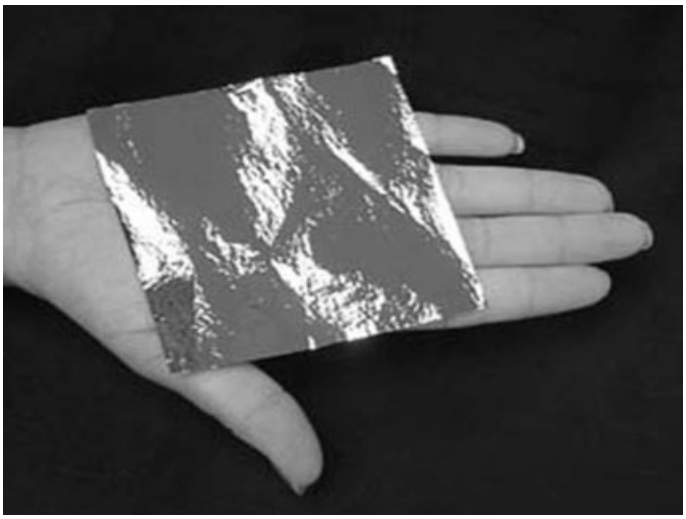


Figure 5.8 Commercial-size (10 cm by 10 cm) Pd–40Cu membrane 5 μm thick and devoid of pin-holes and tears.

the finished goods because they are slightly harder than pure palladium, which is generally considered too soft for fabricating into very thin foil or thin-walled drawn tube. An example of a 5 μm thick rolled foil of Pd–40Cu is shown in Fig. 5.8. This material is commercially produced by continuous rolling, 10 cm wide and up to 100 m or more in length.

When rolling thin foil or drawing thin-walled tube, it is paramount that the starting ingot be closely examined for voids and other defects, and that the surface of the ingot be cleaned to remove any surface imperfections before starting the rolling or drawing process. Starting with a high quality ingot will ensure a quality final product.

5.4 Membrane Module Design and Construction

Unlike polymeric membrane module designs, which include tube-and-shell, plate-and-frame, and spiral wound configurations, dense-metal membrane module designs commonly use only tube-and-shell and plate-and-frame (also called stacked planar). There is no analogous design to the polymeric spiral wound module, most likely because of the difficulty in reliably achieving edge seals that are durable under operating conditions. Success in fabricating one design or the other is partially dependent on the availability of quality membrane in the requisite form, as discussed above.

The stacked planar module design is readily used with any foil-based membrane because foils are available as flat sheet stock. This design is not so readily adapted

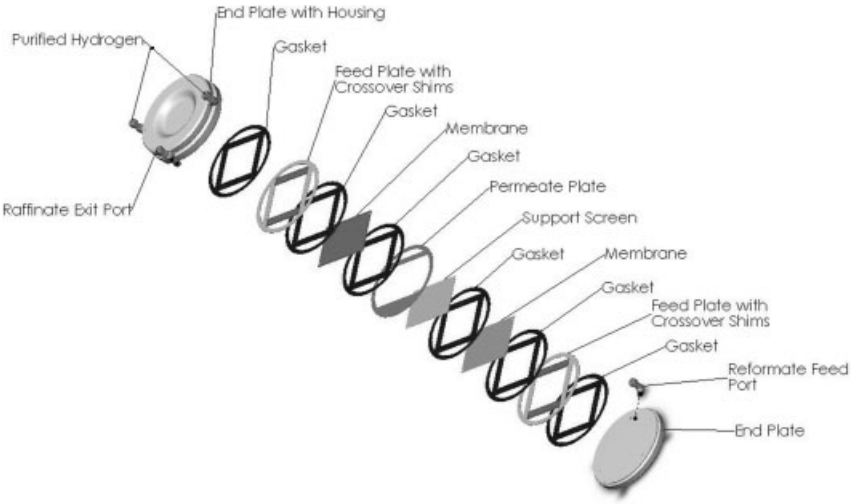


Figure 5.9 Exploded view of a stacked planar membrane array of the type used in the modules shown in Fig. 5.1 and 5.2.

for use with membranes that are based on a thin, permselective layer that is deposited onto a porous support. A stacked planar module design is shown in Fig. 5.9. The membranes are typically arranged in pairs around a common permeate channel. The permeate channels and the feed channels are defined by internal structural elements, or frames. Gaskets are commonly used to form seals between the membranes and the adjacent feed-channel frame and permeate-channel frame. A photograph of representative internal components is presented in Fig. 5.10. For membranes of any practical size (several square centimeters or larger), it will be necessary to support the membrane against the transmembrane pressure differential. The reason membranes are normally arranged in pairs is to

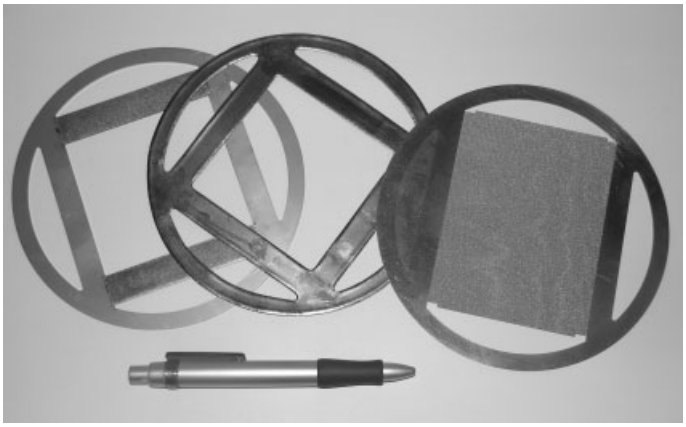


Figure 5.10 Internal module components used in a planar membrane module.



Figure 5.11 Cross-section of a stacked array of planar membranes of the type used in the modules shown in Fig. 5.1 and 5.2.

reduce the number of membrane supports and permeate-channel frames by using one set for two membranes, and to balance the feed pressure so that there is no flexural force applied to the membrane pair. The result is that the membrane pair is only subjected to a compressive force.

Stacked planar membrane modules are relatively easy to build with no requirement for specialized welding or brazing techniques. The membrane modules also offer high packing density because the spacing between membranes may be as small as 1 mm. Scale-up is achieved by stacking more membranes into the module. This type of planar membrane module is easy to fabricate and offers excellent flexibility and reliability. Figure 5.11 shows a cross-section of a module with repeating stacked layers.

The tube-and-shell, or tubular, membrane module is easily adapted for use with drawn tubular membranes as well as membranes that are made by depositing a thin permselective metal layer onto a porous tube support. There are three significant variants to this module design. One is based on the membrane tubes fixed to a header at each end of the membrane tube. The second is similar in that both ends of the membrane tubes are fixed to a header, but to the same header. In the second design, the membrane tubes are bent into a U shape, which can be easily done with small diameter metal tubes. The third is based on a single header, to which open sides of closed-one-ended membrane tubes are fixed. The closed-ends of the membrane tubes are suspended freely. This latter design is more common for commercial applications, due to free thermal expansion and greater membrane durability (see above discussion), whereas laboratory test-and-evaluation practices favor the first variant for its ease of assembly. If the membrane is a drawn, thin-walled tube, the membrane tube will usually be brazed to the header. This is more difficult if the membrane tubes are to be fixed at both ends to head-

ers. If the membrane is a deposited metal layer on a tubular porous support, then gaskets (such as flexible graphite) are normally the sealing method of choice.

Regardless of the membrane module design, effective performance is dependent on rapid mass transfer at the feed side of the membrane. There are several comprehensive references on mass transfer [10], so it will not be addressed in detail here. However, one should consider mass transfer because it affects two design issues: the manifolding of the feed stream to each membrane, and the feed channel design and dimensions. The detailed analysis of these issues differs somewhat when one considers stacked planar modules and tubular modules, but the fundamental objectives are quite similar.

With the stacked planar module, the manifolding of the feed stream may be either internal to the module or external, although internal manifolding is the easier and less expensive option. The manifold may feed each membrane in serial fashion or in parallel, or a combination of series and parallel if there are many membranes in the module. Generally, the only reason to select a parallel feed design is to reduce or minimize pressure drop for the flow of the feed stream through the module. Series flow will give better mixing and better mass transfer of hydrogen to the membrane surface since each flow reversal that occurs as the feed stream flows from one membrane to the next results in mixing of the feed gas. This mixing is beneficial in overcoming the concentration polarization that occurs as hydrogen is removed from the feed stream by permeation through the membrane.

The configuration of the feed channel is also important. If one assumes that the entire feed stream is fed by serial flow through the membrane module, then one can calculate the residence time, or space velocity, for the feed gas as it passes over each membrane. In the worst case, the flow will not be turbulent, and diffusion will be the dominant means by which hydrogen will travel from any point within the feed channel to the membrane surface. Obviously, if hydrogen does not contact the membrane surface, it cannot permeate through the membrane. In this analysis, one needs to estimate the average rate of gas-phase diffusion of hydrogen relative to the residence time that the feed stream is in the feed channel. Cussler [11] has analyzed this type of problem and provides a simple equation to determine if diffusion will be sufficiently rapid:

$$B = l^2/Dt \tag{5.2}$$

Where l is the distance that hydrogen must diffuse to contact the membrane surface, D is the gas-phase diffusion coefficient for hydrogen at the temperature and pressure of operation, and t is the residence time for the feed stream in the feed channel of the membrane. If B is less than 1, then gas phase hydrogen diffusion is relatively fast on the time scale of interest, whereas if B is greater than 1, then the gas phase hydrogen diffusion is too slow.

This is a very powerful means for quickly estimating approximate feed-channel dimensions in order to balance pressure drop with rapid mass transfer. Because the analysis assumes no turbulent flow, it is arguably a worst case scenario. The

membrane module shown in Fig. 5.1 and 5.2B, with the internal structure represented by the exploded diagram in Fig. 5.9, has a feed-flow channel of the following dimensions: 78 mm (wide) × 75 mm (long) by 1.1 mm (deep). The maximum distance that gas phase hydrogen would need to diffuse to reach the membrane surface is half the channel depth or 1 mm. From the maximum flow rate of feed gas (75 L min^{-1}) through the membrane module, the residence time is calculated to be 0.0053 s. The value of D estimated from published tables [12], is about 0.7 cm^2 s^{-1}. Using Eq. (5.2), one calculates the value of B to be 0.82, which is less than 1, and hence can conclude that hydrogen will rapidly diffuse to the membrane surface during the brief time interval that the feed gas is flowing over each membrane.

A general observation of stacked planar membrane modules is that narrow and wide feed channels are preferred to achieve an optimal balance between low pressure drop and good mass transfer, but tubular membrane modules do not allow this configuration for the feed channels. Tubular modules will be of either the floating tube design, in which only one end of the membrane tube is fixed to a header, or a design in which both ends of the membrane tubes are fixed to either a single header or two different headers. When both ends of the membrane tubes are fixed securely to one or two headers, the feed stream will usually be passed through the bore of the membrane tube. As with the above analysis of the feed channels in the stacked planar module, one can use the residence time of the feed gas in the membrane tube, the diameter of the membrane tube, and Eq. (5.2) to estimate if gas phase diffusion of hydrogen will be faster than the residence time for the feed gas in the membrane bore. It does not make sense to flow the feed stream over the outer surface of the membrane tubes because mass transfer will certainly be slow (poor), even if baffles or other means are used to attempt to overcome inherently poor mass transfer. Also, the membrane tube will be more durable if pressurized from within the bore rather than from the exterior.

The type of tubular membrane module, in which the metal membrane tube is fixed at only one end to a header, with the opposite end of the membrane tube closed, offers a unique challenge. With this configuration, it is not possible to have the feed gas flow through the bore of the membrane tube because one end of the tube is closed. Therefore, the feed stream must be passed over the exterior surface of the membrane, and the permeate hydrogen stream is taken from the bore of the membrane tube. This constraint results in poor mass transfer and overall poor performance if the membrane is operated with a mixed-gas feed stream. However, if the feed stream is relatively pure hydrogen, such as would be the case for hydrogen generated from an electrolyzer, then this design is adequate.

With the feed side of the membrane module designed, we will now turn our attention to the permeate side of the membrane. Once hydrogen permeates the thin metal membrane and emerges at the permeate side of the membrane, the pure hydrogen must flow through the membrane support structure and then out of the membrane module. Although there are fewer concerns about mass transfer because the permeate hydrogen is relatively pure, attention must be given to design for minimal pressure drop as the hydrogen flows away from the permeate

side of the membrane, through the membrane support, and exits the module. The membrane support structure is, by necessity, porous, to provide a flow path for the permeate hydrogen. Any pressure drop in the support will be expressed as an increase in back pressure at the permeate side of the membrane with a consequent reduction in driving force for hydrogen permeation through the membrane (assuming a constant feed pressure). (Of course, there must be some pressure drop for there to be flow of permeate hydrogen through the support structure. However, given any fixed delivery pressure of the permeate hydrogen stream exiting the membrane module, the pressure drop through the permeate support equates to an increase (above the exit pressure) in the permeate hydrogen pressure immediately at the permeate side of the membrane.) The effect on hydrogen flux due to high hydrogen partial pressure at the permeate side of the membrane is shown in Fig. 5.12. Ideally, the membrane support structure should be thin and strong, highly porous, and have straight-through pores. The objective is to minimize the resistance to flow of permeate hydrogen through the support.

This is difficult to achieve in practice, but it is always a good practice to comparatively evaluate candidate supports for thickness, porosity, and tortuosity. Support materials that have been used include porous ceramics, porous metal such as sintered metal, metal screens, metal mesh, and slotted metal plates. Metal mesh has been demonstrated to offer favorable performance and attractive cost. The thin Pd–40Cu membranes shown in Fig. 5.4 have been supported on a 70 × 70 mesh; the impressions on the membrane surface are caused by the underlying support. When evaluating the acceptable pore diameter for the membrane support, a general rule is that the pores or openings in the support should not be greater than about four times the membrane thickness. If the membrane is very

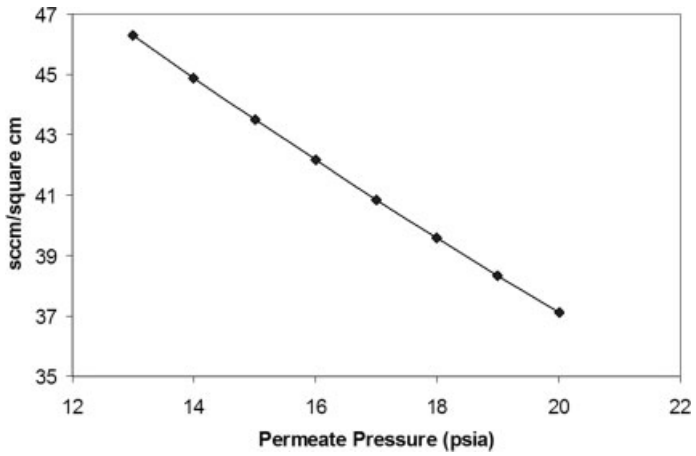


Figure 5.12 Calculated dependence of hydrogen flux on the hydrogen permeate partial pressure for a Pd–40Cu membrane 15 μm thick. The feed is 65% hydrogen at 115 psia (7.93 bara) total pressure; 60% hydrogen recovery (1 bar = 14.5 psi).

thin (<10 μm) and/or is operated at high pressure or temperature, then two times the membrane thickness is a better guideline.

5.4.1 Design of the Module Shell

The membrane module external shell (or housing) is, in essence, a pressure vessel that must be designed to operate at high temperatures, and it must be designed accordingly. In addition, those portions of the membrane module shell that will be in contact with the feed stream must be chemically compatible with the constituents of the feed stream. If the feed is produced by reforming a fossil fuel (e.g., natural gas steam reforming or coal gasification) then the feed will contain hydrogen, carbon dioxide, carbon monoxide, water, various hydrocarbons, and various sulfur compounds (including hydrogen sulfide and organosulfur species). Of these constituents, the sulfur compounds present the greatest challenge to selecting materials that have adequate strength and corrosion resistance. The design engineer should also pay attention to availability of stock material in the alloy of choice, since many exotic and corrosion resistant alloys are not widely available in all standard tube, pipe, and plate forms. Usually, steel alloys with high concentrations of nickel and/or chromium are good selections (e.g., Inconel 600 and 800 series alloys). The common grades of stainless steel, including the 300 series (316 and 304 for example) are not good choices as they are very rapidly corroded by sulfur compounds at temperatures in the range 300 to 600 °C.

Other areas of corrosion concern are the membrane seals. If the seals are composed of flexible graphite, a preferred sealing material as discussed above, then corrosion by the common feed stream constituents is not a problem. However, with any metal seal – whether it be a brazed, welded, or soft-metal gasket seal – it is prudent to evaluate the potential for corrosion specifically at the seal. This evaluation is complicated by the fact that with brazed and welded seals, the metallurgy of the seal is influenced by the addition of the membrane alloy to the liquid braze alloy or weld pool. Experience has shown that welded seals to Pd–40Cu membranes, in which the membrane is welded to either Monel or 304L stainless steel, rapidly fail when subjected to 50 ppm hydrogen sulfide at 400 °C.

For stacked planar membrane modules, the membrane module must be designed so that the amount of elastic deformation (deflection) that occurs during use can be accommodated without compromising the membrane seals. Such deflection occurs as the module is pressurized with the feed stream. Depending on the design of the module, the deflection is likely to be concentrated in the module end plates. The impact of end-plate deflection is minimized by placing the membrane seals as close to the perimeter of the module as possible. Other design features, such as stiffening trusses (Fig. 5.13), may be used to reduce deflection of the end plates [13]. Of course, the degree of deflection is related to the yield strength of the material used to fabricate the end plates (and the module shell in general), so the problem becomes more challenging at higher operating temperatures.

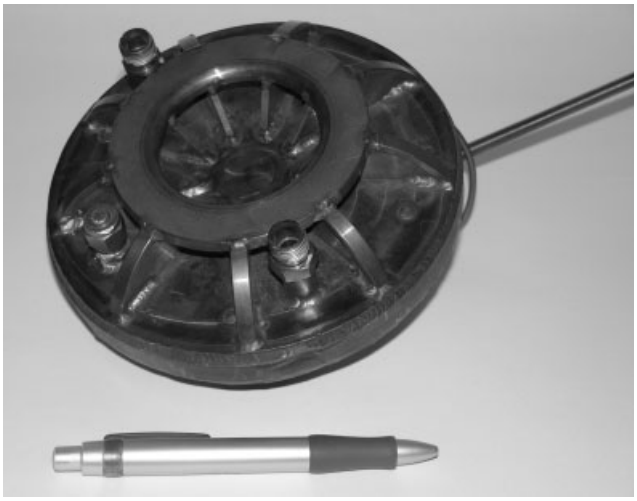


Figure 5.13 Photograph of a truss-stiffened end plate employed on a planar membrane module.

In contrast, tube-and-shell type membrane modules do not have these engineering challenges with deflection and compromise of seal integrity. For this reason, tubular membrane modules will generally be lighter in construction relative to stacked planar membrane modules of comparable membrane area. However, a drawback to the tubular membrane module is that the packing density of the membrane is often substantially less than that which can be achieved with the stacked planar designs.

5.4.2 Membrane Sealing Options

Flexible graphite seals are most commonly used to make seals to the membrane, thereby isolating the feed channel from the permeate channel. The strengths of this sealing choice include versatility, ease of application, and low cost. The major drawback is poor oxidation resistance at temperatures >400 °C. Also, flexible graphite requires a relatively smooth contact surface, and surface roughness or texturing that exceeds 50% of the recommended degree of compression should be avoided. Flexible graphite must be compressed according to the manufacturers' guidelines to minimize leakage, but the leakage will never be absolutely zero as the graphite material has a finite, albeit low, permeability to gases, including hydrogen. For stacked planar membranes, the end plates are used to compress the gaskets. For tubular membranes, some variant of a compression seal using the flexible graphite as the packing material (analogous to packing a valve stem) is the method of choice.

Brazed seals have been the subject of extensive research and development, with only limited success to date. Normally, brazing involves application of elevated

temperature in order to melt the brazing alloy. Furthermore, the brazing alloy must wet the surfaces to be joined; in this case the membrane and the module components that are adjoining the membrane (such as the feed-channel frame and permeate-channel frame for planar membranes, or header for tubular membranes). Once the molten brazing alloy wets the membrane surface, the membrane alloy will be dissolved into the molten brazing alloy. Even if the degree of dissolution is slight, the membrane will be completely dissolved at the brazed seal because there is very little membrane material if the membrane is very thin, perhaps only a few μm to tens of μm thick. Dissolution of the membrane during the process of brazing destroys the integrity of the seal and is not acceptable.

Keys to successful use of brazing are:

- Limit the amount of liquid braze alloy that is in contact with the membrane.
- Join the membrane (for instance, by ultrasonic welding) to a metal frame of copper, nickel, or other metal [14].
- Use a solid-state brazing process.

By ensuring that the amount of metal alloy that is in the membrane exceeds the amount of brazing alloy, one effectively changes the composition of the brazed seal so that it is significantly different from the composition of the original braze alloy, but achieving this objective implies that the brazing alloy be applied in a layer that is significantly thinner than that of the membrane. For instance, if the membrane is 25 μm thick, then the brazing alloy should be less than about 10 μm thick if applied to only one side of the membrane, or less than about 5 μm thick if the brazing alloy is applied to both sides of the membrane to achieve the seal. Not only are special application methods (e.g., plating) needed to apply such thin layers of brazing alloy, but close mating of components to the membrane is also necessary. This all adds cost to the module fabrication.

Despite the potential drawbacks of liquid-phase brazing to seal membranes into the membrane module, one module design lends itself well to this technique: the tubular membrane module incorporating drawn dense-metal tubes. In fact, brazing small diameter closed-one-ended tubes of Pd–23Ag to a header was an early fabrication method used in commercial membrane modules. The brazing process worked well because drawn, tubular membranes tend to be relatively thick walled (>100 μm being common). Thus, the membrane wall is relatively thick compared to the amount of brazing alloy used. However, Pd–23Ag expands substantially (as much as 3%) upon exposure to hydrogen at elevated operating temperatures. This large expansion (and contraction) of the membrane, but not of the brazed seal, frequently leads to failure of the brazed seal joining the membrane tube to the header.

Solid-state brazing is simply a process of solid-state diffusion. Solid-state brazing occurs at a rate which is temperature dependent, meaning that less time is required to achieve a satisfactory solid-state diffusion bond at higher temperatures. To realize diffusion times of the order of a few hours, the temperature should be at least 500 °C, perhaps 600 °C and the parts (membrane and adjacent

module components) must be in direct, intimate contact. There will be no liquid phase present to fill voids or other defects in the joining of the components. Unfortunately, with either liquid-phase brazing or solid-state brazing, it is not possible to determine if the seal will be successful until after the process is completed. At this point, it is not possible to take apart and reassemble the components if the seal has failed – the parts must be scrapped.

Welding is another means of forming a seal between the membrane and the adjacent module components. However, care must be exercised so as not to add too much heat to the weld zone. The weld should be localized as much as possible within the limits of the welding technology. Overheating will result in a large weld pool. This in turn can easily cause the membrane to be melted back from the weld zone (resulting in an immediate failure), or warping of the module components that are being welded to the membrane. Laser welding and micro-TIG (Tungsten Inert Gas) welding are two techniques that have worked well for producing localized welds between 304L stainless steel and Pd–40Cu membranes as well as between Monel and Pd–40Cu membranes. A cross-section of an example weld (micro-TIG) is shown in Fig. 5.14. Application of heat sinks during welding, and appropriate fixturing, are also beneficial steps that are commonly used in welding operations.

Brazing and welding use high temperatures to make seals to the membranes. The temperatures involved vary somewhat, depending on the process, but in almost all cases the temperatures required are greater than the operating tempera-



Figure 5.14 Cross-section of a membrane module showing weld joining of a Pd–40Cu membrane to a Monel module shell. The weld is to the far left of the module. The horizontal line in the middle is where the top and bottom half of the membrane module meet at the membrane.

ture of the membrane module. This leads to thermal stresses that are unique to these fabrication steps. Any differences in the coefficients of thermal expansion of the various materials used in the membrane module, including the membrane itself, will be amplified by high fabrication temperature. The worst case will be high-temperature brazing wherein the membrane module will be exposed to a temperature that may be as high as 700 to 900 °C in order to achieve a brazed seal. The high temperature may strain the membrane (being the weakest material in the membrane module) and cause the membrane to tear or otherwise fail, or to be stretched, as was described above.

5.4.3 Commercial Applicability

Hydrogen is one of the most important industrial and commercial chemicals. It is also the central element of alternative energy technology, because hydrogen is the fuel of choice for fuel cells. Although hydrogen is the most abundant element in the universe, on Earth it is never found in the uncombined, elemental state in nature. Hydrogen must be extracted from other sources, including biomass, hydrocarbons, and water. Since the processes used to extract hydrogen from these feedstocks seldom yield hydrogen of adequate purity, hydrogen separation and purification is both important and essential.

Most hydrogen is used captively; that is, it is made and used at the same site. The largest volume production and consumption of hydrogen is in petroleum refining and petrochemical production. Other applications that use considerable quantities of hydrogen include metal production, glass production, hydrogenation of edible fats and oils, cooling of generators during electrical power production, laboratory instrumentation, propelling space craft, and alternative energy (fuel cells). The commercial or merchant hydrogen market consumed 420 billion cubic meters of gas in 1999. Annual growth rate is steady at about 15%.

Nearly all hydrogen is now produced by steam reforming hydrocarbon or alcohol fuels, partial oxidation of hydrocarbon fuels, autothermal reforming of hydrocarbon fuels, and electrolysis of water. A significant amount of hydrogen is also produced as a by-product of the chloralkali process for making sodium hydroxide and chlorine. A world-scale hydrogen plant based on steam-reforming natural gas or other economical hydrocarbon feedstock will produce 2 million Nm^3 per day of product hydrogen (70 million standard cubic feet per day). At the other end of the spectrum, portable power applications for fuel cells may produce as little as 0.2 L min^{-1} of product hydrogen to generate 15 W of electrical power.

For large-scale industrial applications, the crude hydrogen product stream is usually purified using PSA (pressure-swing adsorption). This is a mature process, but is does not scale down well, and control is somewhat complex given that it is a multi-bed, cyclic operation. Membranes have always enjoyed a preference for small-scale and mobile applications requiring hydrogen purification, and now metal membranes are being reconsidered as a cost competitive alternative to PSA, even for large-scale applications. Because metal membranes operate at elevated

temperature and deliver high-purity hydrogen in a single step (due to their exceptionally high selectivity for hydrogen), the US Department of Energy and several Japanese organizations are currently conducting comparative cost evaluations of hydrogen-selective metal membranes versus PSA for large coal gasification facilities.

The favorable characteristics of metal membranes for hydrogen separation and purification really come to the fore for smaller applications, especially small and mobile applications. For example, compact metal membrane modules are used in laboratory electrolyzers for purifying the product hydrogen to >99.9999%. They are also being used in several fuel cell systems under development ranging from 5 kW combined-heat-and-power systems operating on reformed natural gas, to vehicle auxiliary power systems operating on reformed diesel. In all these examples, the small size and durability of the metal membrane module, combined with the high degree of purity of the product hydrogen, cannot be met by any other hydrogen purification technology. With proper design and selection of materials, metal membrane modules will continue to gain favor and market share in commercial and industrial applications.

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